E-commerce Sales & Customer Analysis

Mohammad Shadab | Sales, customers and order operations

This report examines 12,000 distinct retail orders across seven countries. It uses the repository’s cleaned transaction values and a documented reconciliation of mixed-format dates in the raw file.

Measure	Result
All-status post-discount order value	$1,408,524.91
Distinct orders / customers	12,000 / 3,374
Average order value, all statuses	$117.38
Delivered order value	$776,512.79

Main findings

Electronics and Home & Kitchen account for 46.83% of all-status order value. The category mix explains more about the sales total than order counts alone: Electronics has 1,483 orders and an average order value of $256.38, while Grocery has 1,472 orders and an average of $32.01.

Delivered orders account for 54.82% of records. Cancelled and returned orders represent 9.97% and 8.46%. A snapshot of order status is not enough to conclude that the courier or fulfilment process is failing.

The repeat-customer share is 88.47% when every order status is counted, but 66.82% when only delivered orders are used. The metric is sensitive to its definition and should not be presented as a retention or loyalty score.

After applying explicit date-format rules, the date range is January 2024–June 2026. January–June 2026 order value is 1.93% below the same months of 2025. A full-year comparison with 2026 would be misleading.

Order value is not recognised revenue or profit. All-status totals include cancelled, returned, pending and shipped orders. The dataset has no refund ledger, product costs or accounting recognition rules.

01 / Data and preparation

Document the assumptions before interpreting the totals

Check	Result
Raw file	12,180 rows and 13 columns
Clean file	12,000 rows and 17 columns
Duplicate order IDs	180 duplicate rows removed; clean order IDs are unique
Missing cells in clean file	0
Amounts verified	Sales Amount = Quantity × Unit Price USD; Total Amount applies discount

Existing cleaning steps

The notebook standardises country and payment labels, fills missing discounts with zero and missing cities with Unknown, and fills missing ratings with the median. It converts negative quantities to their absolute values and zero quantities to one, then keeps the first row for each Order ID.

Those are modelling assumptions, not neutral repairs. Among the 12,000 first-occurrence raw records, 96 quantities are negative and 90 are zero. Their original business meaning is undocumented, so converting them may remove adjustments or return information.

The raw file has 6,150 missing customer ratings out of 12,180 records (50.49%). Even after deduplication, 6,056 of the 12,000 source records have no rating. Rating comparisons therefore rely heavily on imputation and are not used here to diagnose product quality.

Date reconciliation

The raw dates use five forms. This report preserves year-first dates, interprets slash-separated day-first dates as DD/MM/YYYY, dashed year-last dates as MM-DD-YYYY, and parses named months explicitly. These rules change 865 dates relative to the clean CSV (7.21%). The resulting range is 1 January 2024–30 June 2026; the stored clean dates instead extend to 6 December 2026.

The format rules are inferred from the raw patterns, including dates with components greater than 12. They are not verified with the original data owner. Original files remain unchanged; Analysis/date_reconciliation.csv records each difference, and the HTML dashboard offers both date views.

02 / Sales over time

Use monthly sums and comparable calendar periods

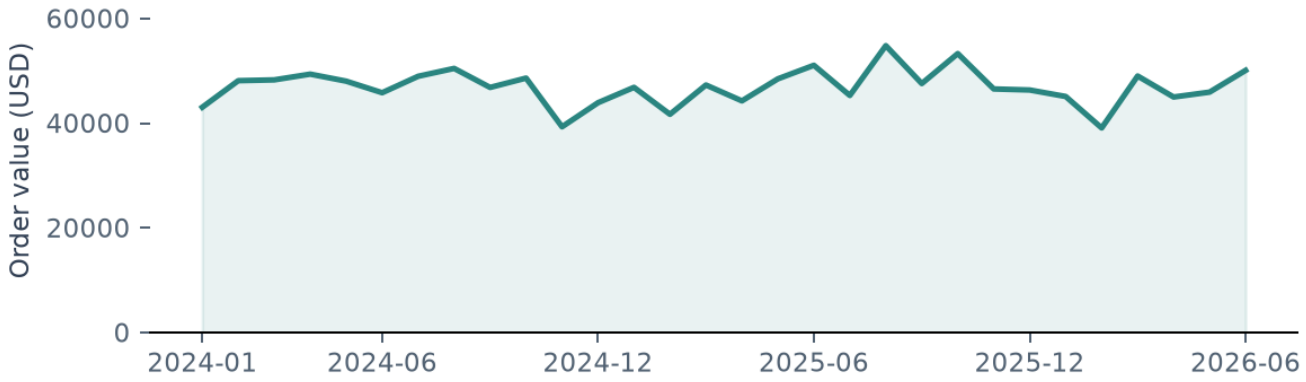


Figure 1. Monthly post-discount order value, all statuses, using reconciled dates. No forecast or seasonal adjustment is applied.

Period	Orders	Order value
2024	4,802	$560,775.39
2025	4,827	$573,518.80
2026 — Jan–Jun only	2,371	$274,230.72

Matched first-half comparison

January–June	Orders	Order value
2024	2,406	$282,672.63
2025	2,374	$279,641.71
2026	2,371	$274,230.72

First-half order value falls from $279,641.71 in 2025 to $274,230.72 in 2026, a 1.93% decrease. Order counts are nearly unchanged: 2,374 versus 2,371. This is a comparison of represented records, not an estimate of underlying business demand.

The notebook’s existing monthly line chart passes transaction-level data to seaborn.lineplot without a sum estimator. Its default aggregation is a mean, even though the chart is labelled as total sales. The revised report and HTML dashboard explicitly sum order value by calendar month.

No increase in Q4 inventory or advertising is justified by this analysis alone. Seasonal conclusions would require a validated date series, consistent coverage and additional years of comparable data.

03 / Products and markets

Concentration in value is different from concentration in order volume

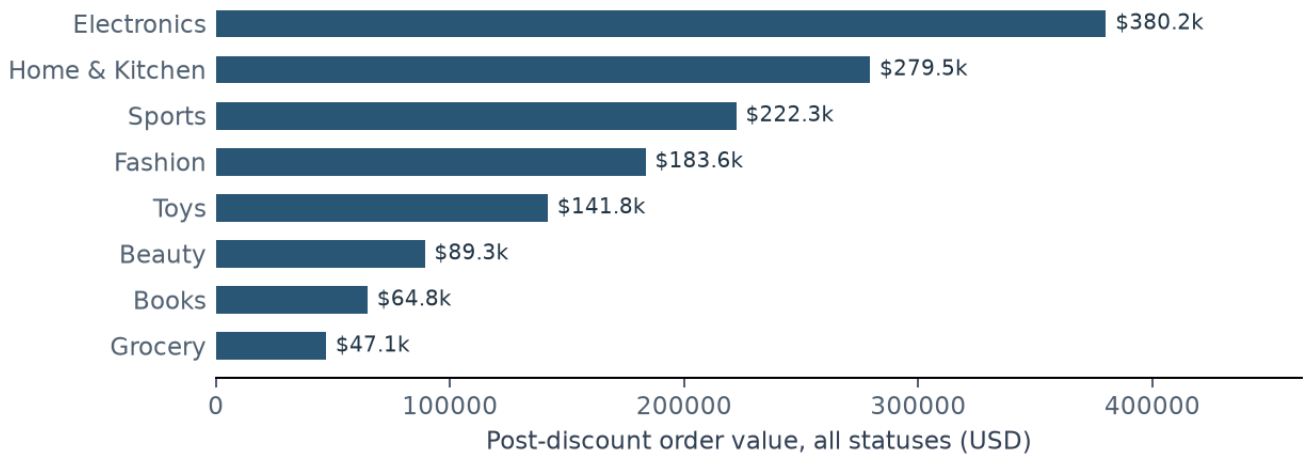


Figure 2. Post-discount order value by category, including all statuses.

Electronics contributes $380,218.59 and Home & Kitchen contributes $279,451.89. Together they account for 46.83% of total order value. These are not profit contributions: acquisition costs, cost of goods and return handling costs are absent.

Market	Orders	Order value
UAE	1,742	$212,693.51
UK	1,768	$206,785.43
Germany	1,747	$203,508.60
India	1,690	$202,527.61
USA	1,711	$200,244.98
Canada	1,635	$193,043.46
Australia	1,707	$189,721.33

The UAE has the highest all-status order value at $212,693.51, followed by the UK at $206,785.44. Abu Dhabi ($108,024.29) and Dubai ($93,903.52) lead the city ranking. City totals include selected order statuses; they should not be mixed with status-filtered SQL results.

Unknown shipping city appears on 494 clean records. It is a missing-location bucket, not a shipping hub. The dashboard excludes it from the leading-city chart but retains those orders in country and overall totals.

04 / Customers and payments

Observed repeat buying is not customer lifetime value

Order scope	Customers	Repeat customers	Repeat share
All statuses	3,374	2,985	88.47%
Excl. cancelled / returned	3,271	2,672	81.69%
Delivered only	2,954	1,974	66.82%

A repeat customer is an ID with more than one distinct order in the chosen scope and observation window. This includes repeated failed or returned orders when all statuses are selected. Comparing the three scopes avoids treating the 88.47% headline as evidence of strong brand loyalty.

Purchase frequency, all statuses

Orders per customer	Customers
1 order	389
2 orders	682
3–4 orders	1,383
5+ orders	920

There are 389 customers with one order, 682 with two, 1,383 with three or four, and 920 with five or more. These counts describe the available window. They do not distinguish new customers from long-standing customers or measure retention between cohorts.

Payment usage, all statuses

Method	Orders	Average order value
Cash On Delivery	2,070	$114.63
Paypal	2,033	$122.57
Credit Card	2,031	$116.62
Netbanking	2,025	$119.96
UPI	1,931	$113.83
Debit Card	1,910	$116.47

The data contains customer IDs but no age, gender or other demographic fields. There is no basis for a demographic analysis. Historical spend in the file is not predicted customer lifetime value; the report does not calculate CLV or cohort retention.

05 / Order status and discounts

Separate operational counts from explanations of why they occur

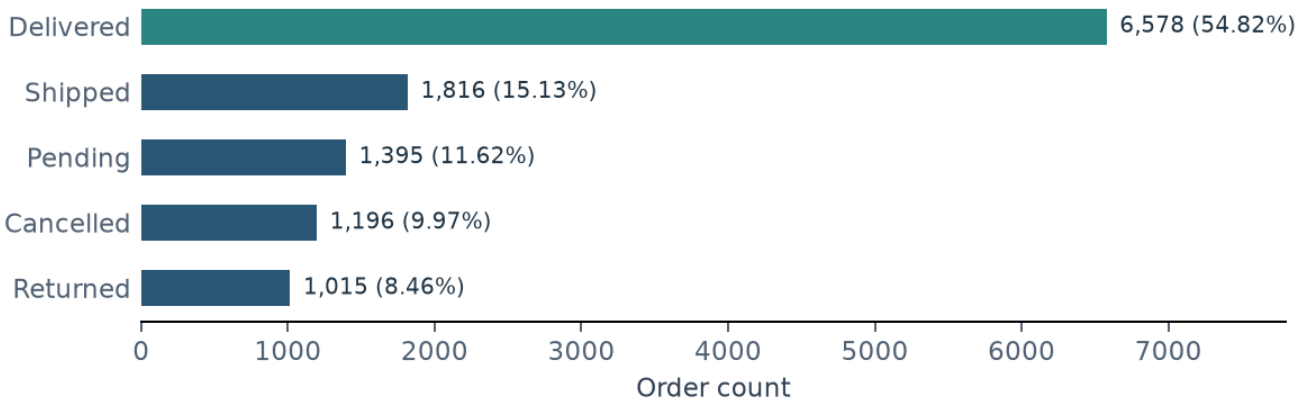


Figure 3. Current order-status distribution across all 12,000 clean records.

Delivered records have $776,512.79 in order value. Excluding cancelled and returned records gives $1,153,844.90, but that subtotal still includes pending and shipped orders. Neither measure is automatically recognised revenue.

The data has no promised delivery dates, actual delivery dates or status-history events. The 54.82% delivered share is therefore a status snapshot, not a measured delivery SLA or a fulfilment conversion funnel.

Discount groups

Discount tier	Orders	Average units	Average order value
0%	3,903	2.08	$129.73
1–5%	1,816	2.01	$116.17
6–15%	3,128	2.05	$110.90
Above 15%	942	2.00	$95.19

This table excludes cancelled and returned orders, matching the intended scope of SQL Q10. Average units per order remain around 2.00–2.08 across tiers, while average order value is lower in the highest-discount tier. Product mix and list prices are not controlled, so these comparisons do not measure price elasticity or a causal discount effect.

Query 3 currently filters “Canceled”, but the data uses “Cancelled”. Cancelled records are therefore retained in that product-ranking output. Correct the status spelling and regenerate the output before comparing it with other status-filtered results.

06 / Definitions and next steps

A consistent measurement basis is the first improvement

Metric	Definition
Gross list value	$1,511,774.11 before discounts; all statuses
Post-discount order value	Sum of Quantity × Unit Price USD × (1 − discount fraction)
Average order value	Post-discount order value / distinct orders in the same scope
Repeat-customer share	IDs with more than one selected order / all selected customer IDs × 100
Status share	Orders with the given status / all orders in the selected group × 100
Observed customer spend	Sum of selected order values per ID; not future lifetime value

Recommended follow-up

1. Validate the raw date formats with the source documentation. Until then, retain both stored and reconciled views and avoid presenting the reconciled range as confirmed ground truth.

2. Standardise order-status scope in SQL, Power BI and reports. Decide whether a measure includes every status, excludes cancellations and returns, or includes delivered orders only; put that choice in the label.

3. Review negative and zero quantities before recalculating sales from raw data. Obtain refund values, delivery timestamps and product costs before making claims about recognised revenue, delivery delays or margin.

4. Evaluate discount campaigns with comparable products and timing, or a controlled experiment. The available comparisons do not support fixed marketing-spend increases, expansion into fulfilment centres or guaranteed business-impact targets.

Calculation and repository notes

Source snapshot: Mohammadshadab1/ecommerce-sales-customer-analysis, commit 5f5f896. Inputs are the existing raw and clean CSVs. A source URL and original data-generation or collection documentation are not included in the repository.

Analysis/build_analysis.py produces the verified metrics and date audit. The revised report is generated from those results. Existing SQL outputs, notebook charts, presentation files and PBIX measures have not been overwritten or certified by this report.

This is a descriptive portfolio analysis. It contains no predictive model, causal estimate, profit calculation or statistically tested operational diagnosis. The supplied records should not be assumed to represent current transactions at a verified retailer.